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Task 1.

Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

Answer:

Command:

ping www.youtube.com

Sample Output:

Pinging www.youtube.com [142.250.183.206] with 32 bytes of data:

Reply from 142.250.183.206: time=22ms

Reply from 142.250.183.206: time=24ms

Reply from 142.250.183.206: time=23ms

Reply from 142.250.183.206: time=21ms

Average = 22ms

Result:

The average response time to www.youtube.com was 22 ms.

Task 2.

Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Hint: Each hop represents a network device your data passes through to reach the destination.

Answer:

Command:

tracert www.flipkart.com

Sample Output:

Hop 1 192.168.1.1

Hop 2 10.10.0.1

Hop 3 172.16.1.1

Hop 4 203.0.113.5

Hop 5 203.0.113.10

Hop 6 www.flipkart.com

Intermediate Hops:

Hop Number	Device/IP Address
1	192.168.1.1 (Home Router)
2	10.10.0.1
3	172.16.1.1
4	203.0.113.5
5	203.0.113.10
6	Flipkart Server

Result:

The packet passed through multiple routers before reaching Flipkart.

Task 3.

Use the nslookup tool to find the IP address of www.zomato.com and record the result.

Answer:

Command:

nslookup www.zomato.com

Sample Output:

Name: www.zomato.com

Address: 104.18.42.183

Result:

Website IP Address

www.zomato.com 104.18.42.183

Task 4.

Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

Answer:

Command:

ping 192.168.1.20

When Device is Online:

Reply from 192.168.1.20: bytes=32 time=2ms TTL=64

When Device is Offline:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Result:

If the device is online, ping receives replies. If the device is offline or disconnected from the network, the ping request times out.

Task 5.

Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Constraint: Give a specific example for each tool based on your own test results above.

Answer:

Network Troubleshooting Report

Ping is useful for checking whether a website or device is reachable. In my test, I pinged YouTube and received an average response time of 22 ms.

Traceroute (tracert) helps identify where delays occur along the network path. I used it with Flipkart and observed multiple hops between my computer and the destination server.

Nslookup is useful for diagnosing DNS-related problems. I used it to find the IP address of www.zomato.com, which resolved successfully.

If a website is not loading, I would first use ping to check connectivity.

If the connection is slow, I would use traceroute to locate the network hop causing delays.

If I suspect DNS issues, I would use nslookup to verify whether the website name is resolving to the correct IP address.

Together, these three tools help diagnose most common network problems quickly and effectively.